

OKELLO ELLY

Software Developer and Systems Architect specializing in scalable systems, full-stack engineering, and cybersecurity.

CONTACT INFORMATION

Location: Nairobi, Kenya

Email: iamellyokello@gmail.com

Phone: +254 769 182 575

LinkedIn: <https://www.linkedin.com/in/okelloelly-102788312/>

GitHub: <https://github.com/Elly739>

Profile

Software Developer and Systems Architect with hands-on experience building secure software systems for student services, transport management, digital learning, analytics, and operational workflows. Skilled in full-stack development, backend engineering, REST API development, database design, networking, and cybersecurity, with authentication experience including JWT and RBAC. Experienced in developing scalable platforms with a focus on performance, maintainability, clean architecture, and secure system design for student services, mobility, and waste management systems. Proficient in AI-assisted development and data-driven engineering, applied in learning, mobility, and waste management projects including LearnLight, Mobi, and SmartWaste.

Education

Bachelor of Science in Computer Software Engineering Zetech University Jan 2025 - Nov 2027

Core Technical Focus:

- Software Development (C, C++, JavaScript, Python)
 - Backend Engineering and REST API Design
 - Database Management (MySQL, SQLite)
 - System Analysis and Architecture
 - Secure Systems and Authentication (JWT, RBAC)
 - Networking Fundamentals and Junior Network Engineering
 - Data Science and Analytics Concepts
-

Skills

- Leadership and team collaboration in student programs and software projects
- Technical communication for project coordination, documentation, and team discussions
- Problem solving and critical thinking in backend development, system design, and analytics tasks

- Strategic thinking and initiative in system architecture, leadership programs, and project execution
-

Work Experience

BIT System - Backend Developer and System Architect

Technologies: Python (Flask), MySQL, SQLite, JWT Authentication, REST APIs, HTML and CSS

Timeline: 2024 - Present

- Designed and implemented a backend platform for student services, event management, payments, and analytics with JWT authentication and role-based access control.
 - Engineered REST APIs with JWT authentication and role-based access control to secure protected endpoints and manage access permissions.
 - Designed normalized relational database schemas to improve query efficiency and support scalable, structured data management.
 - Integrated backend services with administrative dashboards and operational tools to display real-time student, payment, and event data.
-

LearnLight - Software Developer

Technologies: React, JavaScript, APIs, Backend Services. **Timeline:** 2026 - Present

- Contributing to a digital learning platform for structured online education and course delivery.
 - Implemented modular frontend components, backend logic, and API integrations to improve platform scalability and maintainability.
 - Improved performance, usability, and maintainability by organizing the platform with clean architecture and reusable components.
-

Mobi - Software Developer and System Architect

Technologies: React, JavaScript, APIs, Backend Services. **Timeline:** 2026 - Present

- Architecting a multi-modal transport platform that combines public minibuses, ride-hailing services, and campus shuttles in one system.
 - Implementing real-time vehicle tracking, digital payments, route preferences, and commuter trip updates for transport users.
 - Optimized backend services for scalability and reliability, reducing simulated system latency by up to 30 percent.
 - Designed modular system architecture to support expansion to other cities and integration with additional transport providers including minibuses, ride-hailing services, and campus shuttles.
-

SmartWaste - Software Developer

Technologies: Python, Flask, MySQL, REST APIs, Data Analytics Concepts. **Timeline:** 2025 - Present

- Developed a waste management platform for waste collection tracking and operational reporting.
 - Built backend APIs to record, manage, and analyze waste collection data for reporting and analytics.
 - Designed relational database schemas to improve data organization and support operational analytics.
 - Improved reporting and operational decision-making through waste collection data analysis.
-

Certifications

Introduction to Modern AI - Cisco Networking Academy

2026: Foundations of artificial intelligence, AI systems, machine learning concepts, AI applications

Junior Cybersecurity Analyst - Cisco Networking Academy

2025: Threat analysis, vulnerability assessment, security controls, SOC fundamentals

Huawei Computer Networks (CRA Training Program) Certification

2025: Network architecture, TCP and IP, routing and switching, troubleshooting

Cisco Certified Cybersecurity Essentials

2025: Cyber risk management, secure access control, incident response

Data Science and Analytics - HP LIFE

2025: Data analysis, visualization, reporting, analytical thinking

AI Fluency: Framework and Foundations - Anthropic

2025: Prompt engineering, AI workflows, AI-assisted software development

Anticipatory Governance: Systems Thinking and Strategic Foresight for SDG-Aligned Integrated Planning Part 1

2026: Foundations of anticipatory governance, systems thinking, foresight methodologies, SDG-aligned planning

Anticipatory Governance: Systems Thinking and Strategic Foresight for SDG-Aligned Integrated Planning Part 2

2026: Implementation of systems thinking, strategic foresight application, governance integration

SEO - HubSpot Academy

2025: Search optimization, analytics, performance tracking

Content Marketing - HubSpot Academy

2024: Content strategy, audience growth, digital storytelling

Digital Marketing - HubSpot Academy

2024: Campaign strategy, social media analytics, conversion optimization

Leadership and Activities

- Completed the Aspire Leaders Program, Cohort 2 (2026), focused on leadership, systems thinking, and professional development.
- Serving as an Aspire Moderator for Cohort 2 (2026), moderating discussions, supporting participant engagement, and facilitating program sessions.
- Participated in the GMI Mentorship Program (2026), focused on leadership, career growth, and professional development.
- Secretary of the Research and Innovation Club at Zetech University, supporting club events and technology initiatives.
- Led engineering projects and collaborated with technical teams on software development and delivery.
- Pursuing continuous learning in software development, system design, and technical certifications.